

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack platform for structuring **ML training data** (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Built backend security and privacy controls for sensitive customer data, AWS KMS encryption, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**.
- Shipped features rapidly to production as primary technical POC, mentored a second developer, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Shipped a production **LLM** feature (auto-commenting via 4-bit quantized Llama-7B, llama.cpp) with per-persona prompts, async queued generation, and a content-safety filter, built end to end with no ML mentor.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) and migrated core repositories from JavaScript to TypeScript.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform (React) independently end to end; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service that turns unstructured documents and open-ended queries into structured, **source-cited** entity tables via typed retrieval planning, deterministic extraction with LLM fallback, and **cell-level provenance**; hardened with **150+ tests**.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a grounded **RAG** pipeline for multi-hop question answering over documents (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the baseline on a 4,905-question evaluation.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Built a full-stack AI assistant end to end in **TypeScript** that fetches sources and streams grounded research digests over WebSocket, with durable, restart-safe workflows and D1 storage on Cloudflare Workers.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, SQL, MongoDB, PostgreSQL, SQLite

LLM & AI: LLM agents, RAG, structured extraction, document processing, prompt engineering, OpenAI / Groq APIs

Grounding: retrieval (RRF), cross-encoder reranking, MMR, cell-level provenance, source citation, sentence-transformers

Web & Full-Stack: React, Next.js, Node.js, FastAPI, REST APIs, Docker

Cloud & Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.